

1. Administrative & Study Identification

Full Study Title: A Phase 1 Open-Label Pharmacokinetics Study of Palbociclib, a Cyclin-Dependent Kinase 4 and 6 (CDK4/6) Inhibitor, in Postmenopausal Chinese Women with ER (+), HER2 (-) Advanced Breast Cancer **Short Title/Acronym:** Palbociclib PK Study in Chinese Women with ER+/HER2- **ABC Protocol Number:** A5481019 **NCT Number:** NCT02499146 **Sponsor/Company Name:** Pfizer **Investigational Product:** Palbociclib (in combination with Letrozole) **Phase of Development:** Phase 1 **Trial Status:** COMPLETED **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To characterize the single and multiple 125 mg oral dose pharmacokinetics (PK) of palbociclib in postmenopausal Chinese women with ER(+)/HER2(-) advanced breast cancer. **Secondary Objectives:** To evaluate the safety, tolerability, and efficacy of palbociclib plus letrozole as first-line treatment for this patient population. Exploratory objectives include characterizing the longitudinal alterations of pharmacodynamic biomarkers related to the palbociclib mechanism of action (e.g., Ki67, pRb, TK1).

2.2 Background & Rationale As part of the global clinical development program for Palbociclib, studies are planned in cancer patients in China. An assessment of Palbociclib pharmacokinetics in Chinese patients is required by the Chinese Health Authorities to support local regulatory approval. Evaluating safety, efficacy, and cell cycle modulation biomarkers alongside PK ensures comprehensive profiling of the drug in this specific ethnic demographic.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 26 evaluable patients **Number of Sites:** 7 locations across China (e.g., Beijing) **Estimated Study Start:** 11-Sep-2015 **Estimated Primary Completion:** 31-Jul-2018

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Medically confirmed postmenopausal women (spontaneous cessation of regular menses for ≥ 12 consecutive months) or prior bilateral surgical oophorectomy. 2. Documentation of histologically or cytologically confirmed diagnosis of Estrogen Receptor-positive (ER+) adenocarcinoma of the breast and HER2-negative (HER2-) breast cancer. 3. Evidence of locoregionally recurrent or metastatic disease not amenable to resection or radiation therapy with curative intent and for whom chemotherapy is not clinically indicated. 4. Previously untreated with any systemic anti-cancer therapy for their locoregionally recurrent or metastatic ER+ disease. 5. Measurable disease as defined per RECIST v1.1 or bone-only disease.

4.2 Exclusion Criteria 1. HER2-positive tumor as defined by documentation of erbB-2 gene amplification by FISH, CISH, or IHC based on local laboratory results. 2. Patients with advanced, symptomatic, visceral spread, that are at risk of life-threatening complications in the short term (including massive uncontrolled effusions, pulmonary lymphangitis, and over 50% liver involvement). 3. Known hypersensitivity to palbociclib, letrozole, or their excipients.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Palbociclib 125 mg once daily administered on a 3/1 schedule (3 weeks on treatment followed by 1 week off) plus Letrozole (Trade Name: Femara; ATC Code: L02BG04) 2.5 mg once daily continuously. **Route of Administration:** Oral **Duration of Treatment:** Treatment continues in 28-day cycles until disease progression, unacceptable toxicity, or patient withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Routine supportive care and symptom management medications. **Prohibited:** Other systemic anti-cancer therapies, concurrent investigational agents, and strong CYP3A inhibitors/inducers.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Baseline Assessment (RECIST v1.1), Labs
Baseline	Day 0	First Dose, Skin Biopsies for Biomarkers
Interim	Cycle 1 (Days 1-15)	Intensive PK blood sampling (120 hours post-dose for single/multiple doses), Safety Labs, Biomarker tracking
End of Study	Varies	Final Tumor Assessment, Exit Interview, IP Accountability

(Note: Routine clinical visits take place on Day 1 of every subsequent 28-day cycle)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Single-dose and multiple-dose Pharmacokinetic (PK) parameters for Palbociclib including: * Maximum Plasma Concentration (C_{max}) * Time to Reach Maximum Plasma Concentration (T_{max}) * Area Under the Plasma Concentration Versus Time Curve From Time 0 to Time 10 Hours (AUC_{10}) * Apparent clearance, area under the curve (AUC_{τ}), and half-life ($t_{1/2}$) **Measurement Method:** Central Laboratory Analysis using validated bioanalytical assays on blood samples collected predose and up to 120 hours post-dose.

7.2 Secondary & Exploratory Endpoints * **Safety:** Number of participants with Treatment-Emergent Adverse Events (AEs), Serious Adverse Events (SAEs), and Laboratory Test Abnormalities. Incidence and severity graded by Maximum NCI CTCAE version 4.0 Grade. * **Efficacy:** Objective Response Rate (ORR), Duration of Response (DoR), and Progression-Free Survival (PFS) evaluated per RECIST v1.1. * **Biomarkers:** Longitudinal alterations in skin Ki67, phosphorylated retinoblastoma protein (pRb), and serum thymidine kinase (TK1) activity.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: AEs are monitored continuously and graded by NCI CTCAE v4.0. Treatment-emergent AEs are defined as those with an initial onset or increasing severity on or after the first dose. **Serious Adverse Events (SAEs):** Reported within 24 hours (e.g., life-threatening events, hospitalization, persistent disability). **Data Monitoring Committee (DMC):** Internal safety review and monitoring by the Sponsor (Pfizer).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: PK Concentration Population, Safety Population (all patients receiving ≥ 1 dose), and Intent-to-Treat (ITT) Set for efficacy analysis. **Sample Size Justification:** Target enrollment of $N=26$ provides adequate power to characterize the primary PK parameters and steady-state exposure profiles in the target population. **Handling of Missing Data:** Standard non-compartmental analysis (NCA) techniques for PK parameters; missing safety data will not be imputed.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring visits to ensure protocol adherence, accurate RECIST v1.1 tumor

evaluations, and proper intensive PK sampling. **Audit Trail:** eCRF locking, query resolution via data clarification forms, and rigorous chain-of-custody tracking for central PK/biomarker samples.

11. Study Identifiers & Governance

Primary ID: A5481019 **Registry Number:** NCT02499146 **Posting ID:**

855793302196989965 **Sponsor/Lead Organization:** Pfizer **Trial**

Status: COMPLETED **Study Title:** Palbociclib Pharmacokinetics Study In Postmenopausal Chinese Women With ER (+), HER2 (-) Advanced

Breast Cancer **Official Regulatory Title:** A PHASE 1 OPEN-LABEL PHARMACOKINETICS STUDY OF PALBOCICLIB, A CYCLIN-DEPENDENT KINASE 4 AND 6 (CDK4/6) INHIBITOR, IN POSTMENOPAUSAL CHINESE WOMEN WITH ER (+), HER2 (-) ADVANCED BREAST CANCER

12. Clinical Framework (Oncology Specific)

Primary Condition / Disease Area: Advanced Breast Cancer (Metastatic/Locoregionally Recurrent) **Disease Definition:**

Advanced Breast Cancer refers to breast cancer that has spread beyond the breast to involve regional lymph nodes, the chest wall, or the skin (locally advanced, Stage III), or has disseminated to distant organs such as the bones, lungs, liver, or brain (metastatic, Stage IV). This stage of cancer is generally more challenging to treat and, while often incurable, is manageable with various systemic therapies aimed at controlling disease progression and improving patient quality of life. **Diagnosis Threshold:** ER-positive (+) and HER2-negative (-)

Preferred UMLS Name: Estrogen receptor positive, HER2 negative breast carcinoma **Semantic Type:** Neoplastic Process **MeSH Code:**

D001948 **SNOMED ID:** 408540003 **SNOMED Hierarchy:** 254836000 |

Carcinoma of breast | Carcinoma of breast (disorder) **Medical**

Methodology: Targeted CDK4/6 Inhibition combined with Endocrine Therapy (Aromatase Inhibitor) **Optimization Target:** First-line systemic disease control and PK characterization

13. Study Procedures & Methodology

Management Pathway: First-line systemic anti-cancer clinical

pathway **Education Protocol (HCP):** Site initiation visits covering intensive PK sampling protocols and RECIST v1.1 criteria training

Education Protocol (Patient): Informed consent, dosing schedule

education (3 weeks on, 1 week off), and AE reporting guidelines

Quality Audit Mechanism: Central laboratory quality control for PK/biomarker samples (Ki67, pRb, TK1) and independent radiological review if required.

14. Observation & Follow-Up Schedule

Total Duration: Follow-up continuously until study completion (Actual study completion: 24-Dec-2024) **Onsite Visit Cadence:** Day 1 of every 28-day cycle, plus intensive sampling days during Cycle 1 **Remote Monitoring Frequency:** Periodic safety follow-ups as needed between cycles **Checklist Review Interval:** Tumor assessments (e.g., CT/MRI) performed every 8-12 weeks until disease progression

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				